

RISHI MUKHERJEE

Senior Analytics Engineer | Data Engineering & Analytics Platforms

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Summary

Senior Analytics Engineer with 9+ years of experience across data engineering, analytics engineering, and data platforms. Strong hands-on expertise in Apache Airflow, Python, SQL, dbt, AWS, GCP, Redshift, and modern ETL/ELT architectures, with a focus on reliable data pipelines, orchestration, and scalable data workflows. Experienced in pipeline optimization, troubleshooting, data modelling, and translating technical requirements into production-ready solutions. Certified in Apache Airflow 3 DAG Authoring, with strong experience collaborating across Engineering, Product, and business teams.

EXPERIENCE

Senior Analytics Engineer | *Juniper Square*

Feb 2026 - Present

- Owned development of trusted **data pipelines, analytics datasets, and semantic models** powering product, business, and leadership reporting using **Apache Airflow, SQL, dbt, and Looker**.
- Designed and maintained reusable **Airflow DAGs, SQL transformations, and dbt models** supporting staging, transformation, semantic, and reporting layers across complex business domains.
- Developed advanced SQL transformations and reusable analytical models for funnel, cohort, segmentation, and trend analysis, with emphasis on correctness, maintainability, and query performance.
- Troubleshoot **Airflow DAG failures, pipeline issues, data-quality problems, and unexpected downstream behaviour** by analysing task execution, logs, dependencies, transformation logic, and warehouse data to identify root causes and implement corrective actions.
- Optimized **data pipelines and workflow performance** by redesigning transformation logic, consolidating fragmented workflows, and improving underlying data models, resulting in more reliable and efficient downstream data consumption.
- Partnered with **Engineering, Product, and business stakeholders** to gather technical requirements, understand downstream data needs, define acceptance criteria, troubleshoot implementation issues, and translate requirements into reliable production data solutions.
- Improved performance and reliability of high-usage analytics workflows by redesigning underlying data models and consolidating fragmented reporting logic, reducing dashboard refresh times and improving consistency of downstream analytics.
- Established analytics validation and data-quality checks for model and reporting changes, validating expected outputs and identifying issues before production adoption.
- Built AI-assisted analytics and governance workflows to automate **SQL/LookML review, data-model validation, field-reference checks, and data-quality analysis**, improving reliability of production analytics workflows.

Data Analyst | *Jiva Ag*

Aug 2022 – Aug 2025

- Designed and maintained ELT data pipelines integrating operational, product, customer, and marketing data using **Airbyte** and **Fivetran**, supporting reliable ingestion into downstream analytical data platforms.
- Developed and maintained production **Apache Airflow DAGs using Astronomer Cosmos** to orchestrate dbt transformations, data-quality **validation**, and **downstream analytical workflows**, implementing **scheduling, task dependencies, retries, and failure handling** to support reliable data pipeline execution.
- Built reusable dbt models and SQL transformations across staging, intermediate, and analytical layers, applying dimensional modeling, incremental transformation patterns, data tests, documentation, and modular development practices within ELT workflows.
- Monitored and troubleshoot data pipeline and Airflow **workflow issues** by investigating task **execution, ingestion failures, transformation errors, data-quality discrepancies**, and downstream data issues to identify root causes and implement corrective actions; configured **Slack alerts and notifications** for workflow and pipeline failures across Prod and Non-Prod environments.
- Developed **Python and SQL workflows** for data preparation, transformation, validation, statistical analysis, and recurring analytical processes, creating repeatable approaches to data-intensive problems.
- Used Docker and Kubernetes-based execution environments, including Kubernetes-based Airflow workloads, to run and manage containerized data-processing tasks across development and production environments.
- Analysed **customer lifetime value (CLV), cohort performance, campaign effectiveness, and user engagement** to support data-driven decisions around acquisition, activation, and retention.
- Built and maintained recurring dashboards and analytical datasets that enabled stakeholders to monitor **conversion, engagement, retention, campaign performance, and other growth KPIs**.

- Partnered with cross-functional stakeholders to translate business and product questions into **measurable analytical requirements, success criteria, and actionable recommendations**.
- Created technical documentation covering **data sources, transformation logic, pipeline dependencies, data definitions, and validation requirements**, supporting knowledge transfer and consistent implementation across teams.

Data Insights Analyst II | *MoneyGram International*

Feb 2021 – Jul 2022

- Developed KPI frameworks, analytical datasets, and executive dashboards using a **GCP hosted data warehouse**, supporting Customer Analytics, Marketing Analytics, and business reporting initiatives.
- Created and optimized complex SQL queries in **GCP** to drive advanced **customer segmentation, campaign performance analyses**, and loyalty and subscriptions program insights.
- Built **dbt** dimensional data models supporting 100+ **business metrics** and reducing dashboard query latency by **40%** and reduced pipeline runtime by **35%**.
- Created **user stories** and analytical specifications with clear **acceptance criteria**, instrumentation requirements, and success metrics to guide Engineering implementation.
- Identified growth areas through understanding business processes, focusing on Marketing's ad-hoc analyses including **campaign performance, loyalty programs, and migration rates**.
- Built cohort- and segment-based analyses of customer behavior, tracking retention, churn, and lifetime value (LTV) to inform loyalty program and campaign strategy, and supported A/B test design and statistical interpretation for marketing experiments.
- Trained in **Looker Studio BI tool** built performance reporting and BAU dashboards using **LookML** and presented to internal stakeholders.
- Designed and implemented analytical data models within **GCP environments**, supporting scalable reporting solutions, automated data pipelines, and dashboard delivery using **Fivetran** and BI platform
- Conducted end-user training to enable interaction with data using the Looker BI platform, primarily supporting Customer Analytics and Marketing Analytics to drive business growth.

Data Application Analyst | *Harvest Financial Services, Dublin, Ireland*

Sep 2020 – Dec 2020

- Gathered data from multiple applications and posted cleaned, transformed data as data feeds into the system.
- Independently extracted and developed reports for operations and reporting teams by constructing SQL stored procedures, functions, and **SSRS** reports under the reporting lifecycle.
- Used **Power BI** to visualize monthly reporting of asset valuation and customer growth for Harvest Financial Services.
- Developed Master distribution reports using Advanced Excel and SQL for funds liquidity and distributions.
- Built domain knowledge of Irish finance including ARF, PRF, AMRF, Pensions, Ledger, and Investments.

System Engineer | *Tata Consultancy Services, Pune, India*

Jun 2016 – Jul 2019

- Led a cross-functional analytics team responsible for data modelling, data acquisition, reporting, and product analytics, managing the end-to-end analytics lifecycle from requirements gathering to production delivery.
- Developed interactive KPI dashboards using **Tableau, SQL, and Python**, enabling global monitoring of **SLA/OLA** performance and operational metrics.
- Performed ad hoc and exploratory data analysis using **Python** (Pandas, NumPy, Matplotlib, Scikit-learn) and SQL to identify trends, measure business performance, and support strategic decision-making.
- Created **Tableau** dashboards for consumer banking executive reporting; extracted ad-hoc reports from SQL databases and developed RDBMS environments using SQL, T-SQL, and PL/SQL.
- Designed process maps, data flow diagrams, and report specifications using **MS Visio** to communicate project requirements.
- Developed and automated ETL workflows using **SQL Server Integration Services (SSIS)** to ingest, transform, and validate data from multiple enterprise systems.
- Created and maintained database objects (tables, views, stored procedures) using **T-SQL**, writing efficient queries and stored procedures for SSRS reports.
- Created data flow diagrams, process documentation, and reporting specifications to align business requirements with scalable data architecture and analytics solutions.

SKILLS

Data Engineering & Orchestration: Apache Airflow, Airflow DAGs, Astronomer Cosmos, MWAA, ETL, ELT, Data Pipelines, Workflow Orchestration, Pipeline Monitoring

Data Transformation: dbt, SQL, Python, Data Modeling, Data Quality, Testing

Cloud Platforms: GCP Bigquery, AWS Redshift, AWS Glue, AWS Lambda, S3, EC2, AWS Athena

Data Integration: Airbyte, Fivetran, SSIS, Hevo

Cloud: AWS, Amazon Redshift, GCP, BigQuery

Infrastructure & DevOps: Docker, Kubernetes, Git, CI/CD, Terraform*

Data Analysis & BI: Tableau, Power BI, Looker, LookML, Advanced Excel, IBM SPSS, RapidMiner, R Studio, XLMiner, Solver, Google Analytics, QlikView, Google Data Studio, Clevertap, Mixpanel, Metabase, GSuite

Product & Growth Analytics: Product Analytics, Growth Analytics, Funnel Analysis, Cohort Analysis, User Segmentation, Activation, Retention, Conversion, Engagement Analytics, Customer Lifecycle Analysis, KPI Definition, Root-Cause Analysis, Experimentation, A/B Testing, Statistical Analysis, Metric Design

Analytics & Data Science: Pandas, NumPy, Matplotlib, Scikit-learn, Statistical Modeling, Forecasting, Machine Learning, Data Mining

Databases: SQL Server, Oracle 11g, PostgreSQL

Methodologies & Tools: Agile Scrum, JIRA, Eclipse, Visual Studio, Jupyter Notebook, MS Visio, JSON, XML, SSIS/SSRS, SAP Business Objects, Trello

Version Control & ITSM: Git, Tortoise SVN, GitLab, ServiceNow, BMC Remedy

Core Strengths: Stakeholder management, cross-functional collaboration, analytical thinking, project management, communication & presentation, time management, continuous improvement, Product & Engineering Partnership, Requirements Gathering, Success Criteria, UAT, Data Validation, Documentation, Stakeholder Communication

EDUCATION

MSc in Business Analytics

National University of Ireland, Galway, Ireland • Sept 2019 – Aug 2020

Result: 2:1

B.Tech in Electronics and Telecommunication

Institute of Technical Education & Research, Bhubaneswar, Odisha, India • Jul 2012 – May 2016

Result: 72.8% – First Class with Honours

ACADEMIC PROJECTS

- **End-to-End AWS Data & Analytics Platform:** Designed and implemented an end-to-end cloud data platform on **AWS**, covering data ingestion, transformation, analytical modelling, and serving. Built automated data workflows and pipelines, developed a **Cube semantic layer** for governed metrics and reusable analytical models, and exposed analytical capabilities through **FastAPI** services.
- **Predictive Analytics** in the Tourism and Hospitality sector in Ireland using Python, R, Tableau, Exploratory.io, and Power BI.
- **Ethereum price prediction** using an ARIMA model.
- Sentiment analysis using advanced NLP in Python via web scraping data from TripAdvisor, and creation of a content-based recommendation engine distinct from the existing system.
- **Customer profiling and market basket segmentation** based on average order value, average order frequency, demographics, psychographics, and geography, using live POS and CX survey data from a European retailer, in Python and Excel.

AWARDS & ACHIEVEMENTS

- **On the Spot Award**, FY2016–17 — for excellent client communication and completing tasks with high quality and strong performance.
- **Star of the Month Award**, FY2016–17 — for outstanding contribution to a project that improved existing forecasting, increasing yearly profits by 7%.

CERTIFICATIONS, VOLUNTEERING & INTERNSHIPS

- Google Analytics for Advanced Certification
- Scrum Foundation Professional Certificate, CertProf
- freeCodeCamp.org - Data Analysis with Python, certified
- Microsoft Certified: Azure Data Scientist
- Astronomer: [Apache Airflow 3 Fundamentals](#), [Astronomer Certification DAG Authoring for Apache Airflow 3](#)
- Data Engineer Associate, Data Camp
- Member, Business Analytics Society and NUI Entrepreneur Society at NUI Galway